Spirit Lamp

Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Spirit Lamp

What it is used for

A spirit lamp burns an approved liquid fuel to provide a small flame for laboratory heating or demonstrations. It must be used only under teacher supervision with suitable fire precautions.

Main parts

- Identify the fuel reservoir or lamp body.
- Locate the wick and its adjustable holder, if provided.
- Identify the wick opening and flame area.
- Locate the cap used to extinguish and cover the lamp.
- Identify the stable, heat-resistant base.
- Locate suitable ignition equipment and approved fire-control materials.

Before you start

1. Use the spirit lamp only when the teacher authorises the activity.
2. Check the lamp body, wick holder and cap for damage or leakage.
3. Use only the fuel specified for the lamp and laboratory procedure.
4. Fill the reservoir only to the recommended level while the lamp is cool and extinguished.
5. Wipe away any spilled fuel and allow vapours to disperse before lighting.
6. Place the lamp on a stable, heat-resistant surface away from paper and chemicals.
7. Tie back long hair and wear suitable eye protection and laboratory clothing.

How to use

1. Confirm that the wick is fitted correctly and is not excessively long.
2. Ensure the outside of the lamp is clean and dry.
3. Use an approved dropper or filling method to add the specified fuel if needed.
4. Close the fuel container and move it away from the lighting area.
5. Ignite the wick using an approved long-reach ignition tool.
6. Adjust the flame only if the lamp design permits safe adjustment.
7. Keep laboratory materials at a safe distance while heating.
8. Extinguish the flame by placing the designated cap over the wick.
9. Confirm the flame is completely extinguished before moving the lamp.
10. Allow the lamp to cool fully before refilling or storing it.

⚠ Safety

- Never refill a lit or hot spirit lamp.
- Never blow on the flame or attempt to extinguish it with water.
- Keep fuel containers tightly closed and away from ignition sources.
- Never leave a burning lamp unattended or carry it while lit.
- Do not use a leaking lamp or a damaged wick holder.
- If fuel ignites outside the lamp, alert the teacher and use the approved fire response procedure.

Care & storage

- Allow the lamp to cool before cleaning or handling the wick.
- Wipe away fuel residue and inspect the reservoir for cracks or leaks.
- Replace a damaged or heavily contaminated wick as instructed.
- Store the empty or approved fuel-filled lamp according to laboratory rules, away from heat and unauthorised access.

Common mistakes to avoid

- Overfilling the reservoir or spilling fuel near the flame area.
- Refilling the lamp while it is hot or still burning.
- Leaving the lamp uncovered or unattended while lit.
- Using an unsuitable fuel or a damaged wick assembly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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